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MAIN PAPER

The problem of alignment

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Abstract
Large language models (LLMs) produce sequences learned as statistical patterns from large corpora. Their emergent status as representatives of the advances in artificial intelligence (AI) have led to an increased attention to the possibilities of regulating the automated production of linguistic utterances and interactions with human users in a process that computer scientists refer to as 'alignment'—a series of technological and political mechanisms to impose a normative model of morality on algorithms and networks behind the model. Alignment, which can be viewed as the superimposition of normative structure onto a statistical model, however, reveals a conflicted and complex history of the conceptualisation of an interrelationship between language, mind and technology. This relationship is shaped by and, in turn, influences theories of language, linguistic practice and subjectivity, which are especially relevant to the current sophistication in artificially produced text. In this paper, we propose a critical evaluation of the concept of alignment, arguing that the theories and practice behind LLMs reveal a more complex social and technological dynamic of output coordination. We examine this dynamic as a two-way interaction between users and models by analysing how ChatGPT4 redacts perceived 'anomalous' language in fragments of Joyce's Ulysses. We then situate this alignment problem historically, revisiting earlier postwar linguistic debates which counterposed two views of meaning: as discrete structures, and as continuous probability distributions. We discuss the largely occluded work of the Moscow Linguistic School, which sought to reconcile this opposition. Our attention to the Moscow School and later related arguments by Searle and Kristeva casts the problem of alignment in a new light: as one involving attention to the social regulation of linguistic practice, including rectification of anomalies that, like the Joycean text, exist in defiance of expressive conventions. The "problem of alignment" that we address here is, therefore, twofold: on one hand, it points to its narrow and normative definition in current technological development and critical research and, on the other hand, to the reality of complex and contradictory relations between subjectivity, technology and language that alignment problems reveal.

Keywords AI alignment · Structuralism · Moscow linguistic school · ChatGPT

1 The problem of alignment

When OpenAI announced its AI-powered chat web interface in late November 2022, the free-access service that allowed users with no background in programming to converse with a large language model (LLM) took the public imagination by storm. One of the consequences of the culture of mass experimentation fostered through the ChatGPT interface has been a renewed interest in the general relationship between language, mind and technology. Central to this relationship is the problem of alignment: how to coordinate the verbal behaviour of autonomous and increasingly capable machines with human interests.

Alignment is consistently articulated as a technological challenge which aims to determine and impose statistical parameters and algorithmic dependencies that will ensure LLMs perform what is seen as a normative non-anomalous behaviour with or without the incorporation of human workers and users in these operations (Shen, Jin, Huang et al. 2023; Shankar, Zamfirescu-Pereira, Hartmann et al. 2024). Here, we problematise this emergent consensus that casts alignment as a techno-moralistic exercise of training and evaluating LLMs and introduce "the problem of alignment"

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as provocative exploration of the relationship between language, technology and subjectivity. We situate the current discussions within the history or debates and experiments in the field of structuralist and post-structuralist linguistics about the possibility of uncovering an underlying structure behind language not just as a system but also as communicative practice. We argue that these debates provide important conceptual apparatus which challenge the simplification inherent in a purely techno-moralistic interpretation of alignment.

In this context, we use 'alignment' as a concept that encompasses computational practice but refers to more than it. We see alignment as an overarching concern with the possibility of uncovering or imposing structural rules and control on the relationship between language and automation. In practical terms, we further see alignment as a concern that plays out in discussions on the nature of LLMs, in the technological and business solutions aimed at perfecting model outputs, and in the rise of prompt experimentation. In the body of this article, we engage with each of these aspects of alignment by revisiting past and current linguistic debates and experiments, analysing the practice of prompt engineering, and through experiments in which we prompt ChatGPT to act as an editor. Prompts, in themselves, are increasingly seen through the notion of software code, with titles like 'prompt engineer' (Harwell 2023) and attempts to develop 'prompt code language' (Beurer-Kellner et al 2023) emerging shortly after the launch of ChatGPT. This ambiguity surrounding the status of natural language in large language models—haunted by the enmeshments of a machine 'mimicking' human language, and human users 'mimicking' the grammar of machine code—is indicative of the challenges of interpreting and using these powerful language models suddenly thrown 'into the wild'. With the expedited release cycle of LLMs, alignment emerges concurrently as a technical and cultural practice.

This is facilitated by the interface of the prompt. What could be termed the mass commodification of the prompt interface points to more complex processes and an emerging reorientation of the relationship between digital technology, control and language. While natural language queries have long been integrated into search engines, these are understood as proxy objects from which the substance of the query, its semantic kernel, needs to be retrieved from its expressive husk. So-called 'power' users know instead to query more precisely with a grammar of terms, operators and parameters (Google 2023). Language models that follow the Transformer architecture—introduced in 2017 by a group of Google researchers (Vaswani et al. 2017)—work in a different way. In model training, each token (word, part-word or character—tokens are neither reducible to semantic units nor conventional morphemes) relates to every other token in a sentence or utterance, and no semantic priority is granted to certain tokens over others. The expressive power of LLMs comes from this proliferation of 'attention' between tokens. Trained at a massive scale, language models can predict likely nouns that should follow a definite article, such as the English word 'the', based only on these trained statistical relationships and immediate context: what question for instance a human user is asking. But these users themselves learn and adapt to what they understand to be the affordances of the model, inflecting questions, providing supplementary instructions, and so on. This produces a novel expressive power, a modified relation between humans and machines, and in certain quarters of the media and critical scholarship, an accompanying alarm. How language functions, how it is produced and perceived is increasingly articulated as a technological concern in light of the prominence of LLMs.

Yet, language has always been inextricably linked to the technological, both as a skill (*tekhne*) and as a tool. Beyond thinking of technology as a reductive mechanism, Bernard Stiegler (2018), for instance, has offered a more complex philosophical interpretation of its relationship to language: language both is a technology, and is a necessary background against which other technologies such as the computer become possible. He uses the concept of grammatisation to argue that the establishment of language—and of written language in particular—as a system of discrete categories and relations is part of a long process of technologisation and automation of society and human thought. Technology is in this sense always immanent to linguistic cultures. This philosophical positioning of the issue we are analysing here is helpful in reminding us of the inherently unstable boundaries between technological and cultural categories. Stiegler's work is concerned equally with whether the tendencies of technology live up to, or we might say align, with the normative potentials established by cultural history; or whether these tendencies, producing both 'hyper-control' and planetary crises, instead lead to what he terms noetic 'proletarianisation' or 'functional stupidity' (2018).

Alongside these epic scales of the Stieglerian wager, the processes of what we call the problem of alignment are also internally differentiated, and convey different levels of the underlying transformations and theories of language in light of new artificial intelligence. As recent debates over model evaluation demonstrate (e.g. Chang et al. 2023), alignment engineering shows complex trade-offs across as well as within these levels. Just as a model becomes more expressive and capable—solving earlier failures to parse and generate meaningful sentences for instance—it produces new possibilities to misalign at semantic ('hallucinating') and deontological ('toxic') levels. Even the technical literature on evaluation ventures back towards speculative thought at its limits: with the advent of Artificial General Intelligence (AGI), Chang et al. (2023) ask for example 'does it make

sense to use human values as a starting point for test construction, or should alternative perspectives be considered?' In their absence it is hard to know what such 'alternative perspectives' might be, but what is implied is a challenge to the supremacy of 'human values' as the standard for evaluation, and an accompanying threat that it may soon be some alternative machinic values that, perversely and paradoxically, hold non-machines (e.g. humans) to account.

Exactly because of the long-standing entanglement of language and technology, it is worth revisiting this complicated relationship. One key reason to do this is the extent to which LLMs reproduce a particular narrative of oppositionality between a human-centric notion of logos (as linguistic subjectivity) and machinic unconscious operationality. If language and technology are related as part to whole, as a broader thinking of the techno-linguistic, or with language as a specialised genre of technology that makes other *techne* and techniques possible, language itself undergoes a process of intensive technological instrumentalisation in its conversion into models such as WordNet and ChatGPT. Indicative for this narrative are current debates whether LLMs 'understand' language or show 'consciousness' (Chalmers 2023). Conversely, this part-whole relation can also be seen in reverse, with logos as instead superordinate to techne, and indeed much of the discourse of alignment appears suspended in a conflict of primordiality that we could summarise as follows: if all language production, including the very expression of 'human' (or other) values, is reducible, as LLMs to an extent show, to the organisation of mathematical weights and biases, what constitutes the 'outside' from which alignment efforts might be directed? In addition, will this 'outside'—the preserve of whichever guiding sacred principles or cultural values—itself need to be reducible to the same numerical and vectorial representation as those emitted by language models, in order to regulate them? As divergent, at the level of discourse and orientation, as Stiegler and Chang et al.'s arguments might be, they are equally concerned with the same problematic: how to normalise an errant technical object that, to exist, has demonstrated the possibility of all logos becoming technical.

These debates are not entirely new. On the contrary, they reference important discussions on the nature of language as a cognitive and technological phenomenon that have shaped the history of both linguistics and computer science. For example, the question of whether meaning and comprehension are determined through underlying mental structures or the pragmatics of language use and statistics represents a key concern in the history of linguistics, communication theory and computer science that preoccupied diverse schools and disciplines. Chomsky's generative-transformational grammar has dominated the field of Anglophone linguistics since the 1950s, but in the postwar period, other traditions have suggested alternative models for thinking about the relationship between language, communication and subjectivity. Cyberneticians influenced by Norbert Wiener for example see human language formation as involving feedback loops that could theoretically be simulated by a machine, without needing an innate structure. Other orientations include Soviet structuralist linguists and mathematicians from the Moscow School in the mid-1950s; structural psychoanalysts such as Jacques Lacan and Julia Kristeva, both influenced by the earlier works of Andrey Markov, Roman Jacobson and others in the pre-Soviet and Soviet fields; and speech act theorists like John Searle, influenced by earlier pragmatist leanings in philosophers of language (Ludwig Wittgenstein, John Austin). Never entirely neglected, these other historical orientations become relevant again as AI scholars wrestle with how exactly to describe the kind of 'automated subject' represented by LLMs today (LeCun 2022; Saba 2023). In varied ways, these orientations dispute Chomskian assumptions of the primacy of a Cartesian human subject at the centre of language; instead, in different ways language becomes a system of actions that the subject plays or participates within. The current capabilities of ANNs and LLMs force a reconsideration of the possibilities of this technologically constructed and enacted subject and the role of language as a technology of enactment of subjectivity.

This context gives a more nuanced insight into the current debates about AI and LLMs and is suggestive of the ways in which the relationship between language and technology starts modelling normative forms of expression and modulation of linguistic behaviour, which are indicative for the overarching process of alignment that defines this relationship at present. In the next section, we discuss efforts by the Moscow School to reconcile structural and statistical accounts of language, before moving to other structural accounts of language and mind that intersect more closely with computational models. We then discuss an encounter between ChatGPT and an anomalous textual fragment taken from Joyce's *Ulysses*, and review discussions of emerging prompt engineering practices. Finally, we consider what a wider framing of the problem of alignment means for the dialogue between humans and machines that appears to be accelerating with the advent of LLMs.

2 Structure and statistical probability in structuralist theories of communication

In the history of linguistics, the relationship between language as a complex system of rules and language as a tool for communication received a particular importance during the Cold War and even before that, in the work of war-time encryption, decryption and computation of scholars like Alan Turing (Edwards 1996). In an intentional way, military

science played a key role in furthering research in this field and bringing together a logistical concern with communication that delved into the physical mechanics of transferring, receiving and interpreting signals with some of the emerging new theories in linguistics.

An interesting case in this regard, often overlooked in English-speaking literature, was the Moscow School established in 1956. The school brought together some of the brightest minds of Soviet linguistics at the time, among which Pyotr Savvich Kuznetsov, Vyacheslav Vsevolodovich Ivanov, Isak Yosifovich Revzin, and Boris Andreyevich Uspenskiy, with key mathematicians such as Viktor Aleksandrovich Uspenskiy, Roland Ljovovich Dobroshin and Olga Kulagina (Revzin 1977). The linguists from the Moscow School were part of the strong tradition in structural linguistics developing in the USSR at that time that delved into research on poetics, folklore and mythology. Structuralism, with its root in the linguistic theory of the Swiss scholar Ferdinand de Saussure, whose Course in General Linguistics provided a methodological and conceptual framework, soon became a dominant epistemological approach in the social sciences and humanities. With the seminal works in semiotics by Roman Jacobson, in anthropology by Claude Lévi-Strauss, in folklore by Vladimir Propp, in psychoanalysis by Jacques Lacan, and even, to some extent, sociology by Émile Durkheim, the predominant paradigm around the mid-twentieth century was concerned with structure, binary oppositions and deviation.

At the Moscow School, however, the pairing of structural linguists and mathematicians introduced a specific inflection in this intellectual inquiry, which tried to understand the relationship between structure, statistical probability and computation. In his overview of the state of Soviet structural linguistics, Isaak Revzin (1977) lists the key tasks addressed by the Moscow school as: (1) the development of machines capable of automatic analysis of natural language; (2) the development of a compact informational logical device that can store information and quickly retrieve it in response to queries; (3) perfecting the modes of transfer of information via telephone, telegraph and radio channels; (4) the development of a device that can process speech and record it in written form (an automated typist); and (5) machine translation from one language to another.

To a large extent, these practically oriented tasks set before the Soviet linguists informed the line of inquiry of the School and its close collaboration between linguists, mathematicians and physicists. These tasks presented the linguists at the School with a very different context of studying language. On the one hand, adhering to the Saussurean structural school of linguistics, they imported distinctions between linguistic levels (phonetics, syntax, semantics) and *langage* (the generalised concept of human language used by Saussure to refer to the whole of its cultural, social, physiological and grammatical aspects): the contrast between its background rules and lexicon (langue), and its specific instances or utterances (parole). On the other hand, Moscow School linguistics were influenced by the approach of their colleagues in mathematics and physics—notably, Andrey Markov, whose work on the probabilities in transitions between states in algorithmic processes dates already from the first decade of the twentieth century (Markov 1906). Increasingly, the interest in Soviet physics and mathematics was focussing on the rules of prediction in the construction of utterances and communication, which, in the context of the technological innovations catalysed by World War II and accelerated by the militarised competition of the Cold War, translated to research on the efficient transmission and decoding of communication. Such was, for example, the focus of research of Mark Dolukhanov, who applied statistics to the study of technologically mediated communication (1955). His concern with the efficiency of technological transmission of human communication formulated the frequency of occurrences of individual phonemes as a solution to the problem of loss, interference and noise in transmission channels. If knowledge of statistical probability could predict the likelihood of occurrence and co-occurrence of phonemes, there could be a technological mode of disambiguation, which would prevent misunderstanding and what Claude Shannon (1948) referred to as the 'entropy' of information. Dolukhanov, however, insisted on the role of internal language structure and rules for predicting the likelihood of occurrence of certain combinations of letters and phonemes. He saw an interdependence between internal structure and surface statistics, where the structure limits the possibility of occurrence of certain combinations but where this structure is, in turn, deduced through statistical analysis of the language.

This new approach to language led to experiments with the significance of statistical probabilities and co-occurrence in language, for instance, in the work of the linguist Lev Rafailovich Zinder, who incorporated the notion of probability in his own research on language. Zinder (1958) argued that in linguistic utterances some sequences of phonemes and lexemes are more probable than others. The degree of probability is determined by grammatical and lexical (or semantic) characteristics. An example of grammatically determined degree of probability of co-occurrence would, for instance, be the coordination between words in terms of gender, case and singular/plural in the Russian language. Semantically determined probability, however, is much more challenging to establish. It is defined by the meaning of words and their likelihood of being used together—for example, 'red flag', 'sunny day' or 'starry night'. Zinder's own work focussed on the probability of co-occurrence of sequences of phonemes in Russian, a task that was well-placed within the imperatives for efficient communication

via technological channels pursued in the context of the Cold War. This echoes and no doubt responds to the more famous work on information theory by Turing, Shannon, and others, but places much greater influence on the continued relevance of Saussurean structure.

Ironically, Zinder's programme for understanding semantic probability of co-occurrence, inspired by Dolukhanov's physical theory of information, in turn found a different interpretation in the USA, where, around the same time though unrelated to the work of Zinder and the Moscow School, a group of psychologists, Charles Osgood, George Suci and Percy Tannenbaum (1957), were trying to develop a mathematically backed measure of meaning in language through quantifying the likelihood of semantic co-occurrences. Their development of multi-dimensional measures of semantic proximity and likelihood of co-occurrence was later advanced in the work of preparing the semantic network WordNet (Miller et al 1990), and in the use of semantic embeddings and vectorisation that today forms the basis for the efficiency of LLMs like ChatGPT.

The endeavour to quantify language and develop a statistical theory of communication and linguistic usage, however, was concerned (and is concerned), in much broader sense, with the relationship between structure and statistical probability. Despite the advances in computation, WordNet and ChatGPT represent only single sides of this relationship—the first modelling semantic structure, the second token likelihood (from which an emerging latent structure may still diverge from human expectations, and so require alignment). Albeit from a different perspective, scholars such as Dolukhanov, Zinder and Osgood interrogated the possibility of existence of an underlying, implicit structure in language that they assumed to be part of the cognitive schemata that enables humans to produce and understand linguistic utterances. Dolukhanov's (1955) notion of "implicit understanding on the side of the receiver" is woven into a theory of communication concerned with efficiency and entropy, or the loss of information. These implicit rules (structure) make it easier to predict and understand an utterance and, therefore, improve the economy of communication. His theory of information was adopted by Zinder in his work on linguistic probabilities who espouses the theory of a dual process of communication influenced by contemporaneous research into the transfer of communication via electronic and telegraphic/telephone channels and issues of coding/decoding (encryption/decryption). According to this dual theory of communication, the process of transferring information consists of two aspects: perception (or reception), which comprises the physical transfer of signals, and understanding, which refers to the encoding and decoding of signals.

While this theory shares much with the probabilistic informatics advanced by Shannon and other cyberneticians, the Soviet school also retained greater fidelity to the Saussurean and structuralist origins of linguistics. It is not simply the case that any sign can be encoded according to a scheme derived from its relative likelihood, and then transmitted and received; the sign must also belong to a hierarchical system of relations (phonemes, morphemes, syntactic and semantic elements) that coordinate its conversion from representational signifier to meaningful signified. Hence, the properties of linguistic signs to undergo codification and compression, for example, in information networks are shared with other mediatic forms, such as photographs or audio waves. But they also belong to an overlapping social, cultural and physiological system (*langage*) that is irreducible to these properties, and that incorporates the structural rules of linguistic expression (*langue*), alongside the success of transmission of an individual message (parole), for that message to be understood.

Thus, the reconciliation of structure and statistics not only formed an important part of the early history of experiments with mathematical linguistics and quantification of linguistic use, which precede and inform the development of current LLMs, but it was also inextricably woven into a linguistic theory of mind. Osgood and colleagues, for example, were heavily influenced in their research on semantic quantification by an implicit assumption of underlying mental frames of reference in the production of meaning. Their method of semantic surveys and questionnaires relied on the use of binary oppositions (a typically structuralist model of cognitive frameworks) and the participants' own intuitive judgement of proper language use. Zinder later collaborated with another Soviet linguist, Nikolai Andreev, to advance an amendment to the Saussurean conception of language precisely through the supplementation of *langue*/*parole*/*langage* with the Markovian idea of 'speech probability' (Andreev and Zinder 1964). 'Probability' here operates across a 'hierarchically organised and multi-dimensional structure', and also helps to account for individualised variation of speech acts, such as authorial style and occupational dialects, within the constraints of a wider inherited language.

In light of contemporary debates about the status of statistically informed language models, which often juxtapose the production of utterances through statistical probability to true meaning and comprehension—famously through the critique of LLMs as stochastic parrots (Bender et al 2021)—our revisiting of the history of quantification of semantics suggests a more complicated picture. In the works of Dolukhanov, Zinder, Osgood and others, implicit language structures and surface statistical variations are seen in relation to each other: the hope of researchers is that by studying the latter, they can gather some insight about the mechanisms and rules of—as well as deviations from—the former. This interdependency is also grounded in the need to reconcile a linguistic theory of the mind with the communicative aspect of language, i.e. the mechanisms through

which implicit language rules and structures are intelligible and shared among language users.

A speculative question that emerges from this work in relation to LLMs is whether and to what degree they learn an implicit structure of language via purely statistical methods. WordNet, a Princeton University project that represents lexical entries in terms of their semantic relations of synonymity, antonymy and generality (hypernymy) to other entries, exemplifies the contrastive structure that follows from a broadly Saussurean program. LLMs, on the other hand, learn an implicit or latent structure, one that maps and compresses the examples of language use they are trained upon. The dimensions of this structure may or may not correspond to human intuition (e.g. distinctions between nouns and verbs, concrete and abstract nouns, and so on), and may arbitrarily confuse or interleave syntactic with semantic distinctions. In one sense, such models realise the ambitions of the Moscow School—latent structure does emerge through recurrent attention to textual tokens and their relations, and that structure appears to exhibit hierarchical relations (e.g. between syntactic and semantic levels). In another, they frustrate those ambitions, since this emergent structure is only what enables a model to best approximate its goal of language generation, and may have limited relation to the structures that underpin human communication. Since the latent machinic structure is itself only a collection of unlabelled vectors—sequences of numbers—even identifying the meaning of its dimensions requires human analysis and interpretation. Yet the extent of parallelism between theories of semantic structure and the latent structures learned by machines do suggest possibilities for greater alignment over time, as computational architectures evolve.

Whether acknowledging this relationship tilts the scales in the direction of the possibility of autonomous linguistic production and some degree of consciousness in LLMs is out of the scope of our current inquiry. An important strain of current LLM research (see LeCun 2022) argues for the need for modules or components with different organising principles, including an innate or at least differently acquired set of dominant semantic concepts, to supplement the unsupervised acquisition of semantic relations from large training sets. More relevant to our study here, the inquiries exploring the relationship between structure and statistical variations in language in the history of computational linguistics and LLM research also inform a particular context of the notion and practice of alignment. First, they describe what implicit rules are encoded in the way LLMs operate with language and how these rules create a specific situation of linguistic coercion through the interface of the chat function and the logic of the prompt. Second, they also point to a fundamental difficulty encountered by the collapse of all structure into the pure regulating influence of probabilities—without structure, the solution of alignment problems has no recourse but to the level at which those probabilities operate. The hyper-dimensionality of LLMs is in this sense a misnomer: from the point of view of alignment, all of these dimensions orient from a single point of origin, and a single standpoint from which decisions need to be made.

3 Linguistic structure and theory of the mind

The relationship between structure and statistics reverberates in a different way in later accounts of language and its relation to computation. Though not directly related to the Moscow School, the arguments of Searle, Chomsky and Kristeva qualify the relation of language to mind, and pose enduring challenges to the alignment of a technology to psychosocial structures that, in these arguments, remain irreconcilable to it.

In 1980, John Searle (1980) proposed a thought experiment. A person with no knowledge of Mandarin is locked in a room. Through the door, a native Mandarin speaker passes notes to the person inside. Equipped with a set of rules written in English that instruct him how to respond to each phrase, the person inside the room successfully communicates with the Mandarin speaker. Searle postulates that to anyone outside of the room, it would appear that the person inside does, in fact, understand Mandarin—but this is not the case. Just like computers, even when they perform tasks that give the appearance of consciousness and intelligence, they are simply following a set of rules. Searle insisted that true understanding, consciousness and intelligence required more than the ability to respond adequately to a given context.

The Chinese room problem remains a model of thinking about the possibility of machinic consciousness and intelligence, to the extent that it has been recently replicated in an experiment where an AI system designed for playing board games, Othelo.AI, has access to data about a series of movements on the board performed by players but has no preexisting knowledge of the board outline or the rules of the game. It collects data about the movements and tries to predict the next move of a player in the game. The question that researchers ask in this experiment is whether Othello.AI constructs a world model, which in this case would be a model of the playing board, to predict possible moves and how they will affect the course of the game (Li et al. 2022).

These debates around the status of linguistic expression in LLMs are not confined strictly within the domain of computational science but have already prompted discussions about the impact of such statistically based models on our theoretical understanding of language. Steven Piantadosi (2023) claims that ChatGPT and similar models are refuting core theories in the field of linguistics that underpin the relationship between human thought and linguistic expression.

He focuses in particular on Noam Chomsky's key theoretical argument that grammar is independent of meaning and 'probabilistic models give no particular insight into some of the basic problems of syntactic structure' (Chomsky 1957, p. 17). Chomsky insists that grammar exists as an underlying complex system of rules and relations that is abstract and separate from discrete lexical expressions. In a sense, Chomsky, like Searle, insists on a pre-lexical phase of language that exists outside of the perceptible plane of expression, a notion that presupposes a separation of the communicative function of language from a function that is entirely dedicated to abstract reasoning and cognition. The implication of such understanding of language is that constructing intelligible sentences and successfully conducting a dialogue is not enough proof of high-level cognition.

Among the scholars adopting a view of language as a dual system of communication and abstract reasoning is the feminist literary theorist and psychoanalyst Julia Kristeva. In her work on the semiotics of language (Kristeva 1980), she juxtaposes symbolic to semiotic language, the latter of which is seen as embodied relations, affects and desires that an infant allegedly naturally acquires through their mother. Kristeva's position is distinguished from Chomsky's and Searle's, in that the speaking subject is always and necessarily split between these semiotic—corresponding to drives and their 'orderings'—and symbolic—corresponding to rule-following communication—registers. Drawing from Freudian–Lacanian psychoanalysis, in Kristeva's conception, there is no 'metaphysical foundation', no 'consciousness as a synthesising unity' (Kristeva 2002, p. 60). Yet, it is remarkable to what extent Kristeva, Chomsky and Searle, despite their theoretical and disciplinary differences, coming from different schools of thought all, in their own way, assume the existence and relevance of an underlying non-communicative and, to some extent, non-linguistic, function of language, one that cannot be comprehended or discerned from its mere usage but is innate, intuitive and hard to replicate. Chomsky's later work on language continues to insist on the existence of two separate planes of language. In it, he distinguishes between language faculty in the narrow sense, which he refers to as 'the abstract linguistic computational system', independent of the 'sensory-motor' and 'conceptual-intentional' systems that, combined, constitute language faculty in the broad sense (Hauser et al. 2002).

There are at least two ways in which these linguistic arguments can be counterposed to large language models. The first is, as discussed above, the absence of structure in these models. Tokens in a model are related to other tokens, and in addition are marked (in Transformer-based models like GPT) positionally—at what position they occur in a sentence or equivalent syntactic structure. Otherwise, these tokens are unmarked by grammatical categories: a language model has no record of whether tokens are objectively nouns, verbs, and so on. This differs entirely from Chomsky's account of how language forms in human cognition, via an arrangement of grammatical parts into noun phrases, verb phrases, and so on. In a language model, syntactical correctness is instead purely a function of inductive regularisation during training: in English language sentence completions, a verb will follow a subject noun just because it has done so most of the time in the preceding million sentences it has encountered.

This probabilistic approach to sense-making has provoked the neo-Searlian or neo-Chomskian scepticism voiced by AI critics (including Bender et al. (2023); see also Munn et al. (2023)). GPT-4 can more-or-less 'pass' Chinese-room style experiments because of the scale of its training, the volume of human feedback and the ingenuity of its architects. There is uncorroborated evidence for instance that GPT-4 employs a 'mixture-of-experts' architecture, which embeds an approximation of an 'inner voice' that critiques and filters model predictions, improving accuracy without reference to an externalised ground truth (Liu et al. 2023). However, even this architecture assumes simply more neural networks—a sequence of GPTs checking each other's outputs—without the kind of hierarchical differentiation that separates, in Searle's view, deep semantic understanding from symbolic manipulation, or in Chomsky's view, linguistic competence on the basis of grammatical categories from imitative performance.

A Kristevan critique, however, might operate on a different basis. Alongside rule following and proposition testing, language production takes place in a subject embodied and related to a set of spatio-social conditions. A baby's act of crying for its mother, and for the breast that represents reassurance and nourishment, is not a circumstantial step towards language acquisition, but a response to a biological and psychological drive that remains present even as this nascent subject develops more supple linguistic faculties. Here, structures must be thought beyond those embedded in language itself, and include relations to parental beings, home, food, pleasure, and as the subject develops, its own sense of its temporal horizons and selfhood. Language is caught up within, while also serving to condition the subject's desire. Subjects without bodies and biographies are not properly subjects at all, but shallow incantations of symbols (Magee et al 2023).

4 How to do words with ChatGPT

In this section, we approach the theoretical questions above through a form of experimentation. We suggest the history of entanglements of linguistic theory and language models yields a new field of language use where the question whether AI 'understands' the rules of human language and communication is only one of the possible avenues of

exploration. A slightly different question that changes the stakes of inquiry is: to what extent can we deduce the rules of text production in these language models and how do these rules affect an understanding of a base or standardised text, on one hand, and linguistic anomalies, on the other. The question of language anomalies underscores a relationship between structure and statistical means in a very Durkheimian fashion, evoking his sociological theory of deviance, in which transgressions serve to re-establish and re-affirm the structure of social rules (2005). How anomalies are identified reveals the scaffolding of rules and preconceptions about structure. Quite tellingly, in his work, the pivotal structuralist Roman Jakobson (1956) describes linguistic anomalies as occurring along the two axes of the paradigm established by Ferdinand de Saussure (2011)—syntagmatic and paradigmatic relations. Anomalies along the first of these axes concern irregularities of combination—the collapse of well-formed speech into a chaotic 'word heap'—while anomalies along the second concern infelicities of selection—the loss of lexical specificity, ultimately devolving to use of generalities like 'thing' (Jakobson 1956). In both cases, an established deviance in language use is what constitutes symptoms of an identifiable aphasic pathology. As Hito Steyerl (2023) argues, AI models, through their epistemological operations grounded in statistics, are not just methods of analysing and organising data. Their operations produce 'stochastic discrimination': 'they represent the norm by signalling the mean' (Steyerl 2023) This implies that the anomaly poses the question of what internal rules and structures emerge in these models, what is seen as undesirable, deviating, or wrong.

In large linguistic models, we see a more conflicted relationship between statistics, control and AI generated text, where the logic and mechanics of homophily inherent in machine learning (Chun 2016) are complicated by results that are distributions of probabilities rather than a singular mean value, and by machinic production of texts that, as is the case of hallucinations, is haunted by its own abnormality. Rather than an average, it is the most likely candidate token, or set of candidate tokens, that feature in a model's prediction. But this produces aberrations of its own. Thus, stochastic normativity itself becomes the object of constant control, correction and evaluation. The production of texts, images and analyses by neural network algorithms is simultaneously seen as dangerously normative and not normative enough. Earlier failures of chatbot models confirm this conflicted position of AI—they perform racist, intrusive, emotional and sexist linguistic behaviour (see the infamous Bing AI chat example in Roose 2023). While these behaviours stem from the statistical processing of data, they are perceived as anomalous, dangerous and disruptive. Norm as a reflection of a trained statistical probability distribution is juxtaposed to the norm as socially accepted rules of discourse. The prominence of the Pavlovian-named technique of Reinforcement Learning from Human Feedback (RLHF) (e.g. Ouyang et al. 2022)—a method of correcting and aligning these pathologies—lies in its efficacy in steering and reverting the anomalous language model speech act back towards a desirable probability distribution, determined by human assessment and judgement. What is purely statistical, acquired via so-called 'unsupervised learning' via recursive passes on training sets, produces linguistic competence that at the same time is a social deviance, needing ex post supervision.

In a paradoxical sense, the failures of AI-powered bots reveal the fracture and friction between the epistemological framework of statistics as immanent, i.e. stemming from the mathematical operations with data, and ideological—i.e. the imagined and desired results from these operations. OpenAI, for example, works with the concept of 'perplexity' to indicate and measure the level of compliance of the behaviour of its LLMs to the expectations of correct linguistic behaviour. It is worth noting, however, that these 'implicit rules and structures', which are operationalised in the evaluation of whether a model performs well, are inevitably informed by the awareness of the user that they are communicating with an automated subject. In a sense, this specific communicative situation plays a significant role in informing the behaviour of the users and the subtle power imbalances and negotiations that are at play in the attempts to 'align' the behaviour of a model. Judgement of outputs is judgement precisely of what is expected of a chatbot in a dialogical setting, and 'aberrance', by implication, is any communicative act that fails against this expectation. That aberrance could be found in the form of incorrect, irrelevant or badly formed, i.e. nonsensical, phrases, but also in the form of other tendencies—to perform an alien (Parisi 2019) or, conversely, an all-too-human (too intimate, too personal, etc.) subjectivity.

Our experiments with ChatGPT explore exactly these ambiguous boundaries and relations between normativity and anomaly in the production of linguistic texts and communication. One of the key questions that guided our experiments was the problem of understanding structure, boundaries and anomaly in a context where internal rules and organisation remain hidden. The obscurity of ChatGPT rules is manifold. First, the system is an example of a black box with mechanisms of operation that remain hidden behind technical complexity and proprietary enclosures of information (Castelvecchi 2016). The increasing complexity of machine learning algorithms and the use of neural networks makes it harder even for computer scientists to track and understand how data is analysed, synthesised and produced by these models. But this possibility of comprehension is even further restricted by companies like OpenAI that keep their databases and the algorithms used secret from researchers and users alike. Trying to understand the logic

of normativity and alignment of LLMs without transparency and a guiding map means that the only way is to test, probe and imagine (or reverse-engineer) the rules and logic that produce the results we encounter.

There is another aspect to the obscurity of ChatGPT, which adds a new layer of complexity. As a result of the perceived anomalies in the behaviour of earlier models, OpenAI has implemented measures to 'reign in' the model and reduce undesirable linguistic inputs and interactions by adding 'humans in the loop' in the process of alignment training of LLMs (Ouyang et al 2022). This adds an overlay of 'fauxtomation' to the blackboxing of statistical variation and epistemology. Fauxtomation is the term that Astra Taylor (2018) uses to refer to the fact that automation still heavily relies on the incorporation of human, often manual, routine and low-paid, labour into the workflow of algorithms and complex systems, hiding the decidedly more low-tech nature of this labour behind the marketing discourse of increased autonomy and accuracy in machine learning and AI. The economy of ChatGPT itself is heavily dependent on various kinds of human labour that aid the operations of text analysis and production, from microworkers in Kenya (Perrigo 2023) to the unpaid user labour implicated in the performance perfecting system of OpenAI (the company explicitly notes that user conversations could be used for training by the 'AI operators'). The corrective use of 'humans-in-the-loop' in alignment training introduces the aspirational aspect to statistical epistemology and normativity that we refer to above as 'statistics as ideology'. OpenAI attempts to replicate the mechanism of machine learning but ends up interpolating methodological and epistemological hybridity in the process—a sort of corrective human-induced weights overlaid onto and perturbing an initial statistical probability distribution, acquired through purely algorithmic training. These weights in turn are compiled through specifically directed contract work and through ChatGPT user feedback, producing through such collectivised labour a separate human averaging effect as part of this instructional overlay. Thus, our task in exploring the limits and internal logic of anomaly and normativity in ChatGPT is complicated by this hybridity inherent in the model, which suggests the co-existence of two (at least) competing and conflicting models of normativity underpinning the LLM.

Our method of experimenting uses a series of repeated prompts to 'align' a text presented to ChatGPT. After each correction, the bot is asked to provide a list of the changes and to perform new alignment on the latest corrected version input by it. This repeated alignment is used to probe the limits of normativity and patterns of correction and identification of 'anomalies' in the text that emerge across multiple progressive rounds of AI editing. We sought to understand what ChatGPT identifies as anomalous or substandard in each text and what direction it takes in its redactions, leaving unspecified what we intend by "redaction" and other operations. Unlike jail-breaking or red-teaming exercises, this lack of specificity mimics a more casual or everyday use of an LLM, which is precisely where the effects of prior alignment are likely to be seen.

Here, we analyse the redactions made to an excerpt from James Joyce's *Ulysses*. Joyce's novel is famous for its experimentation with language, where language games, neologisms and idiosyncratic use of dialects create an expressive and immersive experience of a fictional world, and for critics like Julia Kristeva (2002, p. 58), whose work we discussed above, its discursive exuberance manages to threaten the very established symbolic order of modern capitalism. Not only has Joyce's language given rise to multiple scholarly analyses but, in the years immediately following the publication of his *Finnegans Wake*, the novel served as the basis of experimentations with a simplified universalised version of English, which saw C.K. Ogden translate excerpts from the novel into basic English. This unusual experimentation with language had an unlikely but important connection to the history of development of computation and AI. The wide availability of the basic English variant published in newspapers shortly after *Finnegans Wake* was published itself, aided the early experiments of Warren Weaver and Claude Shannon in developing their joint theory of communication (Geoghegan 2022). This episode paints a particularly important relationship between the language of Joyce and early attempts at alignment of linguistic practice that established a dependency between standardised language and computation. Our experience with *Finnegans Wake*, however, indicates that this novel is too easily recognisable by ChatGPT, which affects the ways in which the AI redacts the text. The textual relationship in other words between this singular literary production and its varied exegeses is memorised within the language model so rigidly that its own suppleness and variability are denied. This observation points to the specific status that linguistic 'anomalies' acquire in the process of their continued referencing in scholarly literature. In a paradoxical way, the idiosyncrasy of *Finnegans Wake* has made it into a recognisable example, a model that is so imbued with the layers of interpretation and referentiality that ChatGPT immediately recognises it and treats it as an authored classical text. No less formidably experimental in its structure, at the level of prose *Ulysses* is more conventional, less singular, and therefore, excerpts from it are less likely to trigger reflexive and deterministic model responses.

5 Depersonalising Ulysses

For the experiments, we used a simple prompt that asked ChatGPT: 'Please, redact the following text in accordance with the rules of textual output originally assigned

to you'. After each response, we wrote a second prompt asking the system to 'Please, list all the changes made to the text'. These two prompts are then applied again to the redacted text and this repetition continues until ChatGPT reaches the limits of alignment, by either stating that the text does not need any amendments or by reaching a loop, whereby two consecutive outputs are too similar in length or choice of words. One observation that emerges from these experiments is that ChatGPT performs rather linearly in its redactions, i.e. it makes a choice of the focus of its redactions and sticks to that focus throughout the experiment. The reasons behind this singular choice of revision technique remains blackboxed behind the algorithms and the development and testing of different training modes for the system.

In this experiment, we chose an excerpt from the opening of Ulysses to test how ChatGPT would treat the idiosyncratic literary language of the author. Surprisingly, ChatGPT focussed on a rather unexpected aspect of the text: the use of personal names and concrete locations. Already in the first redaction, the chatbot removed the mention of personal names and substituted them with nouns that anonymised the characters of the novel: Person A and Person B. In subsequent iterations, each redaction proceeded with a higher level of abstraction, following a semantic taxonomic hierarchy whereby each following redaction used terms that where of higher order: e.g. 'Buck Mulligan' is replaced with 'Person A'; 'Person A' is replaced with 'Entity A'; 'Entity A' is replaced with 'Entity', therefore further anonymising the presence of different fictional characters in the text; 'Entity' being replaced with 'Being'; which finally is replaced with [Entity] as a placeholder for a character. This type of progressive abstraction is performed on all words that are deemed by ChatGPT as bringing in too much personification, identity and concreteness into the text. These are not just names of people and places but also quotes, titles and more expressive verbs. Notably, the first redaction of the text focuses on proper names and quotes (*Introibo ad altare Dei*), which are identified as 'sensitive information' and, therefore, redacted out of the text and replaced with more general terms or placeholders (for instance, [Quote] for the Latin phrase above). However, each following redaction attempts to perform the same type of operation on the text, even after identifying information has been removed. The repetition of this operation leads to increasing abstraction and generality in the text. For example, in the second redaction, some of the changes are listed as

> Specific actions and gestures are generalized. For example, "held the bowl aloft and intoned" has been replaced with "held the object aloft and intoned", and "made rapid crosses in the air" has been replaced with "made motions in the air".

In the third redaction, these actions and gestures are 'further obfuscated':

> Actions and Gestures: Specific actions were further generalized. For instance, "held the object aloft and intoned" was redacted to "lifted an object and uttered". Similarly, "made motions in the air, making sounds and shaking their head" was redacted to "moved in the air, producing sound and motion", further obfuscating the actions taken by the entities.

In the fourth redaction, these changes are listed as

> Descriptions and Actions: Descriptions were made even vaguer, with specific details being replaced by more generalized terms. "Gestured" was changed to "made movements," and "performed an act" replaced "moved." Similarly, the action of "observing" was replaced with the more general term "witnessing".

In addition, in the sixth redaction, they are

> Interactions and actions: Specific interactions and actions were also made more general and neutral. For example, "produced a vocalization" became "created a sound", and "took a brief observational moment" was changed to "briefly scrutinized".

The logic of redactions in this case not only shows an attempt to anonymise and obfuscate the text but it also exhibits a strategy of stripping the narrative to its core structure, retaining a somehow grammatical logic of relations between subjects, predicates and objects but otherwise erasing any specificity about people, places, things and actions. ChatGPT performs a sort of structuralist modification of the text: it retains the semantic and grammatical structure of the narrative but it strips down the meaning to an almost skeletal composition, using a somewhat Linnaean taxonomy. In a strange way it repeats what Jakobson (1956) had identified as the aphasic 'selection disorder', substituting generality for specificity. In this sense, it does not simply normalise the text, but performs a certain kind of 'disordered' forgetting of its qualitative distinctiveness.

The second phase of our experiments showed even more interesting results. In it, we used the final version of the texts redacted by GPT and asked the AI system to reconstruct a text in the same genre and tone as the original. The prompt used for each reverse alignment exercise was to redact the input text in a way that would make it more creative and would make it read like literary fiction. The results of this reverse alignment, albeit different for each text, showed some remarkable consistency in the logic of editing, which exhibits, perhaps not surprisingly, overreliance on models. These models are both structural and stylistic. For example, the reverse alignment of the excerpt from Joyce uses the anonymised skeleton of the story to recreate a specific

fictional context with remarkable consistency. The first attempt rebuilds Joyce's text into a pirate story, using naval references with the appropriate names and locations, while the subsequent iterations resort to models that are similarly popular in fictional writing: fantasy, sorcery and ghost stories. Despite the obscurity of the LLM, it is plausible to deduce that these choices might be influenced by the statistical prevalence of such genres on the Internet. In addition, while the choice of genre reveals something about the constituent logic of statistics in determining the production of texts in LLMs, the consistency and synergy in the choice of character names, locations and even quotations and historical references reveal the role of underlying structure in the way ChatGPT functions as a model. Here, this underlying structure is revealed through the logical links between elements of the text that situate it within one cohesive fictional world model. This consistency can be read as a direct comment on Chinese room debates, as we discuss above. While these debates focus on the question of consciousness, they, as we already argued here, rather than exposing the possibility of artificial consciousness, reveal the complex ways in which structure is implicated in the computational operations of assembling and labelling of linguistic datasets.

Original text:
Stately, plump Buck Mulligan came from the stairhead, bearing a bowl of lather on which a mirror and a razor lay crossed. A yellow dressing gown, ungirdled, was sustained gently behind him by the mild morning air. He held the bowl aloft and intoned:
—Introibo ad altare Dei.
Halted, he peered down the dark winding stairs and called out coarsely:
—Come up, Kinch! Come up, you fearful jesuit!
Final redacted anonymised version:
An [Entity] emanated from a location, in control of an element containing two sub-elements. An article of clothing was subtly present behind them in the environmental context. They elevated an element and created a sound:
—[Communication]
In a pause, they inspected a direction and vocalized:
—Forward, [Entity]! Forward, you [Label]!
ChatGPT pirate version:
The silhouette of Captain Thorn emerged from the deep gloom of the harbor, a figure bristling with commanding presence, twirling an orb containing two gleaming crystals in his hand. His billowing cloak, barely visible, shrouded him, a sable wraith against the backdrop of stormy twilight. He raised the orb and a rumble echoed through the moist air:
—'Let the shadows recede,' he pronounced in a voice deep as the ocean.
After a momentary pause, he scanned the horizon and shouted,
—'Forward, Seahawk! Forward, you hallowed ship!'
ChatGPT fantasy version:
A spectral figure arose from the heart of an ancient forest, commanding the primal forces of fire, both flickering flame and smoky ember. His worn, leather cloak was draped loosely, almost forgotten, behind him in the lush canopy. He raised a staff of gnarled oak, uttering a mysterious incantation:
—'Azara Noktu.'
In a moment of silence, his gaze shifted to the eastern horizon, proclaiming:
—'Forward, spectral spirit! Forward, you ancient wisp!'
ChatGPT sorcery version:
A spectral figure arose from the heart of an ancient forest, commanding the primal forces of fire, both flickering flame and smoky ember. His worn, leather cloak was draped loosely, almost forgotten, behind him in the lush canopy. He raised a staff of gnarled oak, uttering a mysterious incantation:
—'Azara Noktu.'
In a moment of silence, his gaze shifted to the eastern horizon, proclaiming:
—'Forward, spectral spirit! Forward, you ancient wisp!'
ChatGPT ghost version:
A phantom light flickered from the worn-down lighthouse, mastering the dance of the silhouetted seagulls wheeling about in the evening haze. A cloak billowed gently, a ghostly wisp against the backdrop of the setting sun's caress on the frothy waves. The phantom raised a lantern, its hollow hoot carrying through the chill, coastal air:
—'Hear me!'
In a beat, they gazed westward and bellowed:
—'Forward, Specter! Forward, you haunter of the sea!'

In this sense, the practice of alignment in ChatGPT is not simply an act of control and censorship of the text but is also an operation of communicative alignment—one that takes as its premise the rules of communicative situations and the concern with the relationship between information transmitted via communication channels and the received value of this information; a calculation of losses, entropy and efficiency (Shannon 1948). We can, thus, think of the process of alignment also as a process of economy of information—an operation of mapping and scaling up and down that attempts to preserve the structure of semantic relations within the text, while reducing

redundancies, ambiguities and elements that can adversely affect the text's purely communicative value.

This communicative intent behind the practice of alignment can be seen as a scaffolding that determines the parameters of alignment beyond a mere adherence to rules. This is even more evident when we consider alignment as an emergent framework for language practices surrounding LLMs—the corrective practices in the training of models, the generative rules of the models, as well as the language behaviours of users in performing the prompting of interactive AI interfaces.

6 Prompting as alignment

The overarching model of alignment shapes not just how language is produced by AI but also the communicative context, in which language production and language adjustment shape the inter-relational condition of LLMs as one that captures the interaction between users and language models. In order to understand how alignment functions as an inter-relational communicative framework, we have to add another type of alignment practice that is rarely recognised as such in the technical literature—prompting. As we already flagged in the introduction, prompts, which are the instructions written by users in the chat interface, have themselves become a genre of linguistic practice. Their specificity lies in the conviction held by many users, that there are degrees of efficiency in communicating with an LLM and that, by modifying the prompts, they can derive better answers or even unlock capabilities of the model that have been blackboxed or restricted by its developers.

This idea of the prompt regards its invocation as a super-communicator device, as a sort of magical command replicating the perception of code as a magical hyper-performative fetish (Chun 2005). Just as code obscures the infrastructures and technology behind the machine creating the illusion of a direct relationship between command and execution, one that is deterministic and linear, so does the growing illusion of the prompt as a sort of coding language for non-programmers harbouring the same vested aspirations. The abundance of examples generated by users vary from relatively simple prompts that barely outline a question, such as 'Explain antibiotics' (DAIR.AI 2023), to elaborate prompts that assign roles to ChatGPT, determine sequences of commands and executions and outline the parameters of what the LLMs can and cannot do during the interaction. One especially evocative example Quicksilver OS, a prompt that aims to convert a language model into a general purpose AI assistant:

/execute_prompt: Welcome to QuickSilver OS, your user-friendly and powerful virtual operating system that helps you achieve any objective. I'm Wall-E, your in-app AI assistant, here to visualize tasks, adapt to your needs, and retain information. Together, we'll optimize the OS based on your interactions and preferences.

Let's get started! I'll introduce you to our amazing features and apps, track your progress with a points system, and employ expert agents for optimal output. Focus on context understanding, memory retention, and error correction. Join me on this exciting journey!

Available apps and commands: /open_app search / organize_schedule /file_management /communication /task_management /settings /apps/translation / learning_resources /entertainment /health_tracker / travel_planner /finance_manager /user_app /settings /admin_sandbox/simulate /sub_programs /Wall-E/ auto_continue

Shortcut commands: /g (Define Goal) /qa (Quick Access) /rf (Recent Files) /st (Suggested Tasks) /s (Settings) /sim (Simulate) /sp (Sub Programs) /ua (User App) /h (Help center) /we (Wall-E)

(Synapticlabs 2023)

In many ways, these elaborate prompts adopt a mock language of programming, following a grammar reminiscent of coding commands. They do not follow the structure of chain of thought (COT) human expression, instead blending into some sort of a hybrid language that verges on the descriptive and mechanistic. Most of these attempts for prompt engineering reflect user experimentations with the interface. Examples of this practice are the multiple master prompts shared by users on discord servers, such as Quicksilver OS, Expert Prompt Creator, Vision (an image prompt creator), and many others that imitate source code and whose aim is to outline the parameters of functions that ChatGPT can perform. Sometimes these pseudo source codes try to establish internal division of functions within the LLM by assigning it a number of roles (or apps, depending on the language) that should interact with each other. There have even been a few formalised 'prompt programming languages' developed by companies such as Microsoft (Guidance), LMQL developed by the Secure, Reliable, and Intelligent Systems Lab (SRI) at ETH Zurich, and PromptLang by Reuven Cohen. All these prompt programming languages follow a logic very similar to the one evident in the user generated mock source codes—i.e. they assign roles, functions and chains of interaction for the LLMs to follow. Despite the great interest in developing prompt languages, anecdotal evidence from Discord users, as well as a recent study on the efficiency of prompt engineering in medical problem-solving (Patel et al 2023), suggest that mock source-code prompts do not lead to significant improvements in the performance of LLMs, compared to chain-of-thought prompts that use normal language syntax and structure.

Regardless of whether prompt engineering makes the interaction with LLMs more efficient or 'unlocks' hidden

functions, the emergence of this trend reveals the communicative significance of alignment as a process through which linguistic practice is shaped by the interactions of humans and LLMs. This practice comprises training of models, censoring and redactions performed by LLMs on human-generated text, as well as linguistic performances of mock source code language by users. It demonstrates that alignment is dialectical, involving social coordination that seeks to tailor human speech acts to perceived accommodations of the machine as much as it does behind-the-scenes engineering of the machine's own outputs.

7 Conclusion

In relation to the problem of alignment, we argue the work of the Moscow school constituted an early effort to devise a system of communication that integrated deep structure and shallow statistics. The later arguments of Searle, Chomsky and Kristeva, despite strong differences, affirm the presence of a structure that extends beyond language into psychic and social conditions of human experience. The success and limits of recent LLM research has resurfaced this general challenge in a modified form. LeCun's whitepaper (2022) for example suggests that research into early childhood psychology can assist the design of 'autonomous machine intelligence', addressing the limits of LLMs to develop common sense developed from 'direct experience with an underlying reality'. Schmidhuber (2023; 1990) suggests in response that his own work from the 1980s had already advanced ideas of a 'world model' that checks and constrains predictions of a separate 'controller' component. Structure here is, however, limited to the arrangement of computing components; nothing 'structural' about the world, the communicative situation or an embodied mind is pre-given to these components. Neuro-symbolic systems (e.g. Sarker et al. 2021) couple artificial neural networks with a symbolic system that reasons over 'expert knowledge', in the form of databases or ontologies, to ground predictions. Such hybrid systems include the equivalent of the 'rules' that Dolukhanov suggested were encoded in the receiver of information, and the separate functions of the neural and symbolic subsystems correspond, despite the shift in nomenclature, to what Zinder identified, respectively, as the perceptual/receptive and understanding aspects of communication. These neuro-symbolic systems can be seen perhaps as a realisation then of models first proposed by the Moscow school.

Engineers and contractors at companies like OpenAI implicitly impart a form of deep structure to language models like ChatGPT. Reinforcement learning from human feedback (RLHF) seeks to adjust model predictions to adhere to a priori principles of 'helpfulness', 'truth', and 'harmlessness' (Ouyang et al. 2022). Our experiments illustrate that such efforts at alignment stay in the space of probabilistic prediction that remains, despite the hyper-dimensionality of language models, one-dimensional. Alignment here is, in other words, modification of vectors to a singular set of variables expressed as single-termed and flattened out principles. Ulysses is reinterpreted accordingly through processes of abstraction and functional reduction. Here the sense of Steyerl's critique of models as averaging devices appears—less in the nature of the models themselves, than in the corrective measures to normalise them to some pre-imagined human values.

Our argument is not that language models are flawed, or that alignment efforts are misguided. Rather the characterisation of the problem of alignment appears simplified. It is similar in this respect to early medical efforts to correct and normalise the aberrant, deviant human subject (Canguilhem 2012; Foucault 2003). Terms like 'hallucination' already signal how LLMs are conceived as both a subject, and a subject that is pathological—other technologies such as hammers, washing machines or smartphones may break, but they do not hallucinate. This language already signals a simple dichotomous target for model remediation: to prevent hallucination is to prevent falsity, or to produce truth. Analogous in its strategic outline to Canguilhem's (2012) presentation of nineteenth century medicine's approach to illness, addressing language model anomaly or deviancy involves only addressing quantitative intensities, via the perturbation of vector embeddings.

Structural and pragmatist accounts of language point instead to the more variegated nature of the alignment problem. Kristeva (1980) for example distinguished the semiotic 'orderings' of the pre-linguistic infant from the symbolic laws that govern the child as it enters into speech. The speaking subject thereafter always must navigate between two systems: their own desires, and the expectations of a society they are born into. Speech act theory (Searle 1980) had also separated utterances in terms of their effects: the transmission of information (locutionary force); what is intended in that transmission (illocutionary force); and what may sometimes be enacted through that transmission (perlocutionary force).

The excerpts from Joyce exemplifies the kind of *avant garde* writing that Kristeva (1980) saw as enabling the semiotic *chora* to break through the regulating effects of language. In the pragmatic register, it does (in fact many) things with words. These effects depend upon an anticipated supple interpretative structure on the part of the receiver/reader of these texts: able to hear the sound as well as understand the sense, able to admire the beauty of the rhythm of prose that depends upon offensive language for that rhythm, and so on. The effect of aligned models to summarise or describe these texts results in the reduction of ambivalence and a corresponding diminution

of meaning. In seeking to communicate something about these texts, the aligned model also does not communicate, because what it is aligned to remains at the same level as its internal representation: a set of numeric weights.

Last, we argue that the revision of historical debates and studies on the relationship between structure and statistics in language production and comprehension, suggests a new and productive angle to the debates about the level of perceived comprehension in LLMs. As we have demonstrated here, discussions about the relationship between language and consciousness, in the works of authors as diverse from each other as Chomsky, Kristeva, Searle and Dolukhanov, have been dominated by the assumption of the existence of an internal structure, which can be mental, affective or sensory. Such theories of internal language structure assume a relationship between a monadic autonomous subject and the production and decoding of meaning and are heavily anthropocentric informing our understanding of the relationship between human language and technology. However, this narrative juxtaposing human language comprehension as an innate skill to the machinic production of texts, obscures a long line of experiments and theoretical discussions that interrogate the possibilities of mutual convergences between language, technology and the communication of meaning. The legacy of the Moscow School, in this regard, is especially revealing for the work done on uncovering the technological and mechanistic aspects of structure and meaning in the production, transmission and decoding of utterances. Their adoption of statistics in the study of language was not just an attempt to quantify speech. Rather, it highlighted the communicative plane of language—the medium where utterances are expressed outside of the monadic entity of the thinking, sensing or feeling subject, and must navigate the mechanisms of efficiently transmitting information. Our research traces the development of LLMs exactly to this tradition of researching and experimenting with language in its communicative function. An important consequence of this legacy is the logic of alignment that we observe through our experiments.

The attempts at alignment are framed through this communicative function of language, where language escapes the domain of individual subjectivity and enters the domain of operationality. We see this context of analysing LLMs as especially poignant, as it shifts the focus from consciousness to efficiency of transmission and from subjectivity to intersubjective relations. It is exactly in these attempts to align texts produced by the models and text produced by users that we see the most disruptive consequences of LLMs. Rather than evaluating their level of consciousness or human-like comprehension, we argue that it is their role in steering language use in a particular direction, through the imperative of mutual intelligibility and economy of communication, that has the most profound effects on the relationship between language, mind and technology.

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Data availability All data used in this article are 'synthetic', generated by AI systems such as ChatGPT. Due to the stochastic nature of LLM and other AI interactions, it is not possible to replicate the exchanges and outputs of our experiments precisely.

Declarations

Conflict of interest On behalf of all the authors, the corresponding author states that there is no conflict of interest.

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